

Backend Engineer — Take-home Task

Please read carefully before starting. Deadline: 5 days from receipt.

Pick whichever option interests you more — both are weighted equally. Focus on clean, readable code over completeness. Cut scope if needed.

Option A — Café Management MVP

Three modules, loosely coupled:

1. Ordering — manage orders with status tracking: pending → preparing → done
2. Billing — generate invoices with tax and discount logic (flat % tax, flat % or fixed discount)
3. Inventory with AI — accept an image upload, use a vision model (GPT-4o, Gemini, Claude, etc.) to identify the item, and log it to inventory

Example: photo of a coffee bag → identified as "Arabica Beans" → stock updated

Option B — DuoAccent Accent Scoring API

Build a POST /score-accent endpoint that accepts an audio clip and returns a 0–100 score with feedback across four parameters:

1. Pronunciation — accuracy of individual sounds
2. Stress — word and sentence-level stress patterns
3. Liaisons — natural linking of words in connected speech
4. Intonation — rise and fall patterns

Hard constraint: latency must be under 10 seconds (server-side, from request receipt to full JSON response). Document how you met this — or what you would do differently with more time.

What to include (both options)

- Working REST API covering the requirements above
- Input validation and error handling with correct HTTP status codes
- Dockerfile — we should be able to run it with docker compose up
- README — how to run it, your design decisions, what you'd improve with more time

How we review it

- 40%. Design decisions and trade-offs
- 35%. Code clarity and structure
- 25%. Correctness and validation

We are not looking for a production-ready system. We want to see how you think.

Stack

Use whatever you are comfortable with. We work with Python (FastAPI) and Java (Spring Boot) internally.

How to submit

Submit your GitHub link within 5 days using the form below:

<https://forms.gle/C2mgjXt1N8pF1YgL7>

Make sure your repository is public before submitting. We will confirm receipt within 24 hours and follow up within 2 business days.

Questions? Reply to your interview email — we are happy to help.

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